The Chinese University of Hong Kong, Shenzhen

MAT3300 Mathematical Modeling — Final Project Guidelines and Grading Criteria

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Weight	55% of the course grade: report 30 + slides 10 + Q&A 15
Team	1–3 students, free grouping; the whole team receives the same mark
Task	Solve **one** of the three contest problems in `Problems/`
Deliver	Report (PDF) + code + slides, **Sunday, November 29, 24:00**
Q&A	Every team has a 5–10 minute Q&A in class on Dec 8, 10, 15 or 17

1. Overview

A contest-style modeling project: from a real-world question to assumptions, a computable model, and results you can defend, in writing and in code. The **Requirement** section of the official problem statement is the specification of your project. Where its submission rules differ from this document, **this document applies**.

2. The three problems

Choose **exactly one**. The official statements are in the folder `Problems/`.

A 2023 MCM Problem A, *Drought-Stricken Plant Communities*

B 2021 MCM Problem B, *Fighting Wildfires*

C 2021 ICM Problem D, *The Influence of Music*

Notes.

- Read the full official statement before choosing. Its page limit, glossary and extra document apply.
- You may read published solutions for ideas. `References/` contains two Outstanding Winner papers per problem as examples of format and writing, not model answers. Anything you take from any paper must be cited; copying is plagiarism.
- Problems A and B come without data files: take numbers from the statement or public sources, or assume them, and cite every source. Problem C: the four files in `ProblemC_Music_data/` are the only data you may use.
- Where the statement leaves a choice open, make a choice, justify it, and test how the conclusions change when it changes.

3. What to submit

Submit through Blackboard, one submission per team, as three files. List the student IDs of all team members in ascending order.

- `Team_<StudentID1>_<StudentID2>_<StudentID3>_report.pdf`
- `Team_<StudentID1>_<StudentID2>_<StudentID3>_code.zip`
- `Team_<StudentID1>_<StudentID2>_<StudentID3>_slides.pdf` (or `.pptx`)

The report (top of the Summary Sheet), the slides (first slide) and the code `README` must state the full name and student ID of every team member.

(1) **Report**: PDF, **at most 25 pages in total**, in this order: one-page **Summary Sheet**; **Table of Contents**; your complete **solution**; the **extra document** below (a short piece for a non-expert reader); **References**.

- A: none.
- B: a one- to two-page annotated Budget Request to the Victorian State Government.
- C: a one-page document to the ICM Society on the value of your approach and on further study.

Add one final page, not counted in the 25, with a short **contribution statement** (who did what). Teams of one may skip it.

(2) **Code**: all code needed to reproduce every figure and number in the report, with a `README` saying which script produces which result. Include the input data you used.

(3) **Slides**: at most 8 slides summarising the project for a reader who has not read the report.

Deadline: Sunday, November 29, 24:00. Late submissions follow the late policy of the assignments.

4. Grading

Component	Points
Report (including the extra document)	30
Slides	10
Q&A	15
Final project total	**55**

Report (30 points).

Criterion	What we look for
A. Fulfilment of the Requirements	Every item of the problem's Requirement is addressed, correctly and completely.
B. Modeling quality	Clear and justified assumptions; a well-defined, correct model.
C. Validation and critical assessment	Checks of the results, sensitivity analysis, and an honest discussion of limitations.
D. Communication	A Summary Sheet that stands on its own; an extra document suited to its reader (where the problem asks for one); clear organisation, figures and references.
E. Code and reproducibility	Results can be reproduced from the submitted code.

5. Q&A sessions

- Every team has a Q&A session with the instructors. The sessions are **public**: other students may attend. Each team has a **5 to 10 minute** slot in class on one of December 8, 10, 15 and 17. The order of the teams will be announced later on Blackboard. Timing is strict.
- Every member must attend. A member who does not attend without prior approval receives no marks for the Q&A.

- Questions can cover any part of your report, code and slides, and may be directed at specific members. You may need to start with a brief overview of your project, and you may use your submitted slides if needed.

6. Evaluation form for slides and Q&A

The slides (10 points) are judged on criteria 1–4, from the slide deck. The Q&A (15 points) is judged on criteria 5–8, in the Q&A session.

	Criterion	What we look for
Slides		
1	Graphic-based communication skill	Readable figures and tables; slides not overloaded with text.
2	Clear structure with introduction, content and conclusion/perspective	The problem, approach, results and conclusion are easy to identify.
3	Knowledge of subject matter and familiarity with the topic	Correct modeling; assumptions understood and justified.
4	Clear presentation and logic	A clear line from question to model to result to conclusion.
Q&A		
5	Language-based communication skill	Clear, concise answers within the time.
6	Knowledge of subject matter and familiarity with the topic	Real understanding of the model and the results.
7	Familiarity with own work	Every member can explain any part of the report and code.
8	Performance in Q&A	Direct, correct answers; honest about limitations.

7. Academic honesty and use of AI tools

- You may use AI tools freely and need not acknowledge them. You remain responsible for every statement, equation, number and line of code in your submission; understanding is checked in the Q&A.
- Cite every external source of data, ideas, figures and code. Do not invent data, parameters or references.
- Copying between teams or from published papers is plagiarism. The course has zero tolerance for cheating and plagiarism.